

FRESH-MAP SELF-IMAGE EROSION: LABELLED KERNELS AND A FORCED JORDAN BLOCK

ANONYMOUS

ABSTRACT. Fix an integer $n \geq 1$ and repeatedly intersect a labelled subset $A \subseteq [n]$ with its image under a freshly resampled uniform endomap of $[n]$. We determine the transition to every labelled target while retaining the total image size of the sampled map on A . An $(n+1)$ -state cardinality quotient then gives every-time labelled transition and absorption formulas. The full operator has eigenvalues $a!/n^a$ with Boolean multiplicities; the equality of its top two size-layer eigenvalues is not semisimple, and the quotient contains a forced 2-by-2 Jordan block for every $n \geq 2$. Finally, products of a polynomial quotient give the joint image-size marks at arbitrary epochs. Ordinary random-image occupancy, Stirling surjection counts, and generic triangular-chain algebra are treated as background. The literal conjunction remains owner-thin and its external status is `HOLD_EXTERNAL`.

1. THE LITERAL CHAIN AND THEOREM PACKAGE

Fix an integer $n \geq 1$ and write $[n] = \{1, \dots, n\}$. At epoch r , sample an independent uniform map $f_r : [n] \rightarrow [n]$ and update

$$(1) \quad A_r = A_{r-1} \cap f_r(A_{r-1}).$$

Only the restriction of f_r to the current set affects this update. The process therefore differs from the ordinary random-image chain, whose image sizes and compositions already have a direct literature [3]; random mappings and their classical statistics are also mature [1]. Successive-elimination and leader-election games provide another nearby vocabulary [2]. In one simultaneous round of that directed-graph game, selected endpoints are eliminated, so the survivors are zero-indegree vertices. In (1), by contrast, the selected endpoints inside the active set are retained, so they have positive indegree; moreover each fresh map is sampled into the fixed ambient set $[n]$, rather than along the induced current graph. We assign all of those objects, Stirling surjection counts, and generic finite-chain linear algebra no contribution credit. They are included below only to make the literal conjunction self-contained.

Let $\left\{ \begin{smallmatrix} a \\ k \end{smallmatrix} \right\}$ be a Stirling number of the second kind. For $0 \leq b \leq a \leq n$, set

$$(2) \quad H_n(a, b; k) = \binom{n-a}{k-b} k! \left\{ \begin{smallmatrix} a \\ k \end{smallmatrix} \right\},$$

$$(3) \quad N_{ab} = \sum_{k=0}^a H_n(a, b; k),$$

$$(4) \quad Q_{ab} = \binom{a}{b} \frac{N_{ab}}{n^a},$$

where a binomial coefficient outside its natural range is zero. At $a = 0$ the denominator is one. Let P denote the full transition matrix indexed by the 2^n labelled subsets.

Theorem 1 (labelled, spectral, and marked atlas). *For the chain (1), the following statements hold.*

(i) *If $B \subseteq A$, $|A| = a$, $|B| = b$, and $k \geq 0$, then*

$$(5) \quad \#\{f|_A : A \cap f(A) = B, |f(A)| = k\} = H_n(a, b; k).$$

Consequently $P(A, B) = N_{ab}/n^a$; a target outside A has probability zero.

(ii) *For every $t \geq 0$ and $B \subseteq A$,*

$$(6) \quad P^t(A, B) = \frac{(Q^t)_{ab}}{\binom{a}{b}}.$$

This formula is zero when $B \not\subseteq A$.

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(iii) The complete algebraic eigenvalue multiset of P is

$$(7) \quad \left\{ \frac{a!}{n^a} \text{ with multiplicity } \binom{n}{a} : 0 \leq a \leq n \right\}.$$

For $n \geq 2$, the common value

$$(8) \quad \lambda = \frac{(n-1)!}{n^{n-1}} = \frac{n!}{n^n}$$

has one $J_2(\lambda)$ block in the cardinality quotient. Hence the full operator is not diagonalizable. No complete Jordan form for P is claimed.

(iv) For $n \geq 2$, the empty set is the unique recurrent state. If $\tau = \min\{t : A_t = \emptyset\}$ and $|A_0| = a$, then

$$(9) \quad \mathbb{P}_a(\tau \leq t) = (Q^t)_{a0},$$

$$(10) \quad E_0 = 0, \quad E_a = \frac{1 + \sum_{b < a} Q_{ab} E_b}{1 - Q_{aa}} = \mathbb{E}_a \tau.$$

At $n = 1$, both subsets are fixed. At $n = 2$,

$$(11) \quad Q = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \end{pmatrix}, \quad E_2 = 4.$$

(v) Define the polynomial quotient

$$(12) \quad Q_{ab}(z) = \binom{a}{b} n^{-a} \sum_{k=0}^a H_n(a, b; k) z^k.$$

For any $t \geq 1$, the (a, b) entry of $Q(z_1) \cdots Q(z_t)$ is the joint generating polynomial of $|f_r(A_{r-1})|$, $1 \leq r \leq t$, and the final size, summed over all labelled b -targets. Division by $\binom{a}{b}$ gives the polynomial for each fixed $B \subseteq A$.

The last statement retains information lost by the subset endpoint. It does not assert that the epoch marks are independent.

2. ENDPOINT COUNTS AND LABELLED POWERS

Proof of Theorem 1(i). Fix $B \subseteq A$ and require $|f(A)| = k$. The part of the image outside A is a $(k - b)$ -subset R of $[n] \setminus A$, giving $\binom{n-a}{k-b}$ choices. Once R is fixed, the restriction $f|_A$ must map onto the k -set $B \cup R$. There are $k! \binom{a}{k}$ such surjections. This proves (5). Summing over k and dividing by the n^a restrictions gives the stated probability. The values of f outside A would contribute the common factor n^{n-a} if full maps were counted. $\square$

Summing (5) over the $\binom{a}{b}$ possible targets proves $\sum_b Q_{ab} = 1$. Thus Q is the transition matrix of the cardinality process.

Proof of Theorem 1(ii). Every trajectory is nested. Simultaneous relabelling of $[n]$ commutes with the distribution of (1), so conditional on the initial a -set and final cardinality b , all $\binom{a}{b}$ labelled endpoints have equal probability. The total probability of that layer is $(Q^t)_{ab}$, which proves (6), including $t = 0$. $\square$

This argument is stronger than a size marginal: after the small quotient has been powered, it recovers every labelled endpoint exactly.

3. THE FORCED JORDAN BLOCK AND ABSORPTION

Order labelled subsets by nondecreasing cardinality. Since transitions only go to subsets, P is lower triangular. A self-loop at an a -set requires $f|_A$ to be a permutation of A , and therefore

$$(13) \quad P(A, A) = \lambda_a = \frac{a!}{n^a}.$$

This proves the algebraic multiset (7).

Proof of the Jordan assertion. The ratios $\lambda_{a+1}/\lambda_a = (a+1)/n$ are strictly below one for $a < n-1$, while (8) gives the only repeated adjacent quotient diagonal. Also

$$Q_{n,n-1} = \binom{n}{n-1} \frac{(n-1)! \binom{n}{n-1}}{n^n} > 0.$$

Solve $(Q - \lambda I)x = 0$ from row zero upward. The distinct first $n - 1$ diagonal entries force $x_0 = \cdots = x_{n-2} = 0$. Row $n - 1$ is then free, but row n and the displayed positive coefficient force $x_{n-1} = 0$; only x_n remains free. Thus the repeated eigenvalue has algebraic multiplicity two and geometric multiplicity one, giving exactly one J_2 in Q .

Functions of subset cardinality form a P -invariant subspace on which the operator is represented by Q . The nonsimple restriction proves that P itself is not diagonalizable. $\square$

Proof of Theorem 1(iv). For $n \geq 2$, every nonempty proper set has a positive chance of strict loss: one may map all its points to a point outside it. The full set also has a positive chance to enter a proper layer. A finite monotone chain then reaches zero almost surely, and zero is the only recurrent state. Formula (9) is the all-time kernel with final size zero. First-step conditioning gives (10). Direct substitution gives (11). When $n = 1$, the sole endomap fixes both subsets, which is why that boundary is separate. $\square$

4. IMAGE-SIZE MARKS AND THE CLAIM BOUNDARY

Proof of Theorem 1(v). Equation (5), summed over the $\binom{a}{b}$ targets of size b , is exactly (12). Multiplying polynomial transition matrices conditions successively on the intermediate sizes and multiplies the corresponding monomials. Relabelling symmetry at the final layer then gives the fixed-target division, as in (6). $\square$

The owner boundary is deliberately narrow. Random-map statistics and ordinary image-size composition are subtracted through [1, 3]; successive-elimination and leader-election vocabulary is subtracted through [2]; occupancy and surjection formulas are classical; and triangular spectra, Jordan algebra, and hitting-time recursions are generic. The latter game's zero-indegree survivors are the complementary selection convention to our positive-indegree retained set, and its next round uses the induced graph rather than a fresh map into a fixed ambient label set. Internally, earlier random-intersection notes use group translations, graph cuts, or permutation fixed sets. None supplies both the image-occupancy endpoint sum and the top-layer coupling in the literal process (1).

The standalone verifier enumerates every restriction for all subsets through $n = 6$, checks full labelled powers through four epochs, and performs exact rational Jordan tests through $n = 9$. These finite checks target arithmetic, boundary, and labelling errors; they do not prove the displayed formulas. The source search was bounded, and its failure to locate the literal conjunction is not evidence of novelty, priority, or freedom to operate. Accordingly the manuscript remains an anonymous internal note with status `HOLD_EXTERNAL`.

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